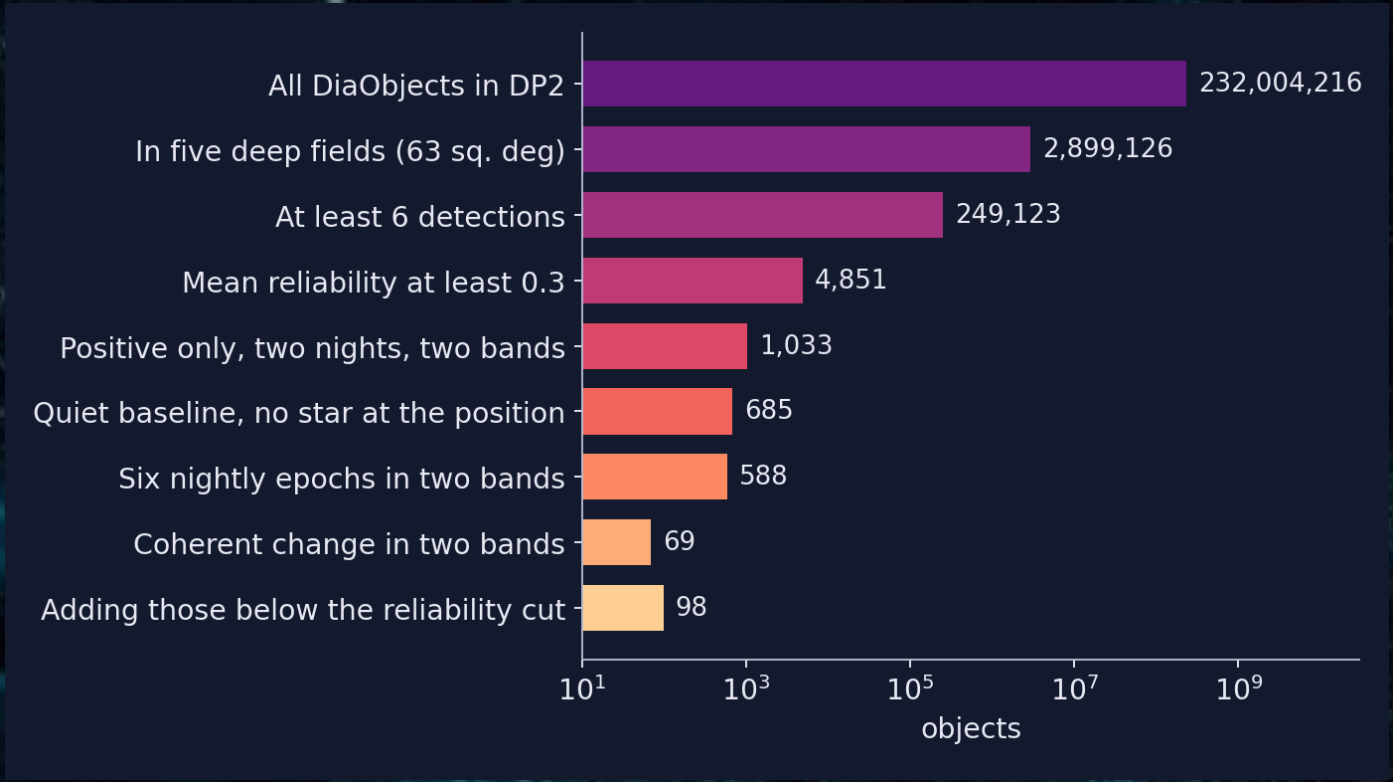
Finding transients in DP2



reliability 0.01

median over objects with six or more detections: the reliability score does most of the work

142 point sources

removed: a persistent star-like source sits at the position in the Object table

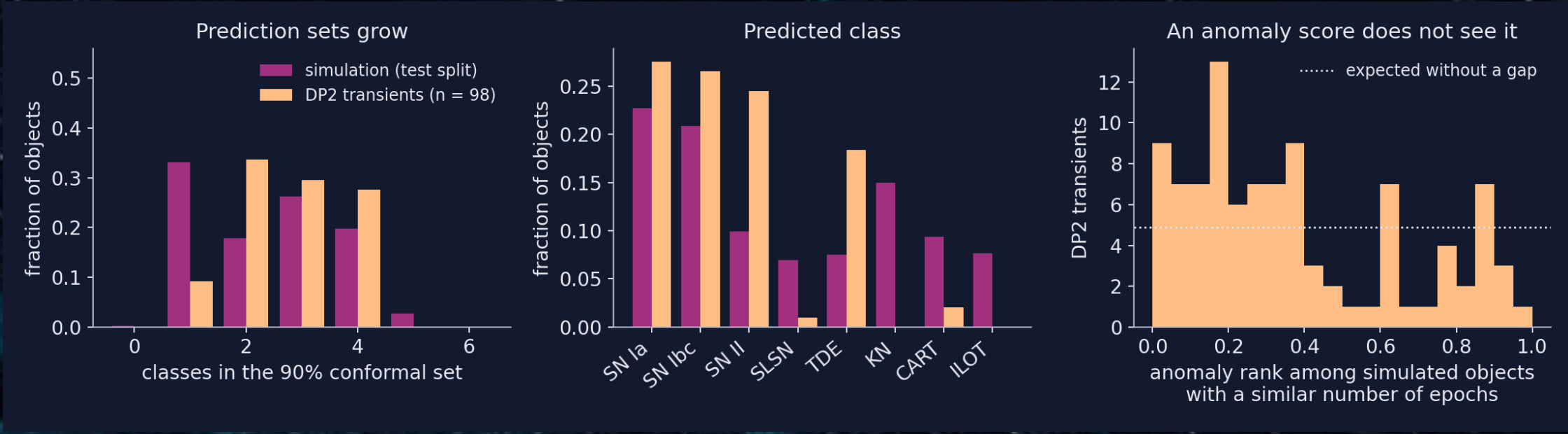
baseline -0.26 sigma

median forced flux away from the event: the template contains transient light

From 232 million DiaObjects to 98 transients that brighten or fade coherently in two bands.

From simulation to sky

Six-band FLARE trained on SASSAFRAS simulations only, applied to 98 real transients from the DP2 deep fields. No labels.



33% to 9%

objects with a single class in the 90% conformal set, simulation to DP2 (9 of 98)

AUC 0.99

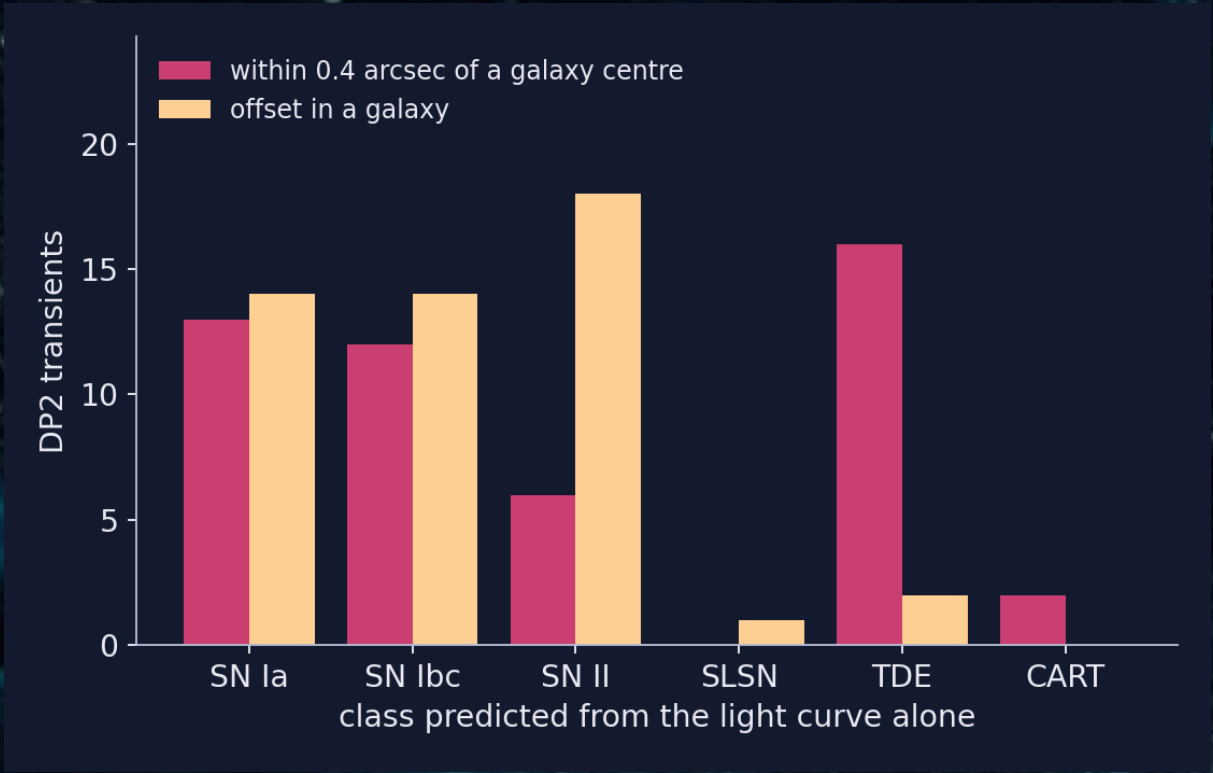
a classifier separates simulated from real feature vectors (0.50 for simulation against simulation)

median rank 0.31

real transients sit inside the simulated cloud: the anomaly rank stays low

The model is unsure on real data, and its prediction sets say so.

A check the model never saw: the host galaxy



16 of 18 TDE calls

sit at a galaxy centre

18 of 24 SN II calls

are offset from the centre (SN Ia: 14 of 27)

65% same class

per visit against nightly light curves: cadence changes the answer

71 of 75 confident calls

were TDE before cleaning: faint, flat, nuclear, probably AGN

Predicted classes follow the host position, which the model was never given.

Proposal: a white paper from this school

What

what we learned applying machine learning to DP2: one or two pages per hackathon project

Who

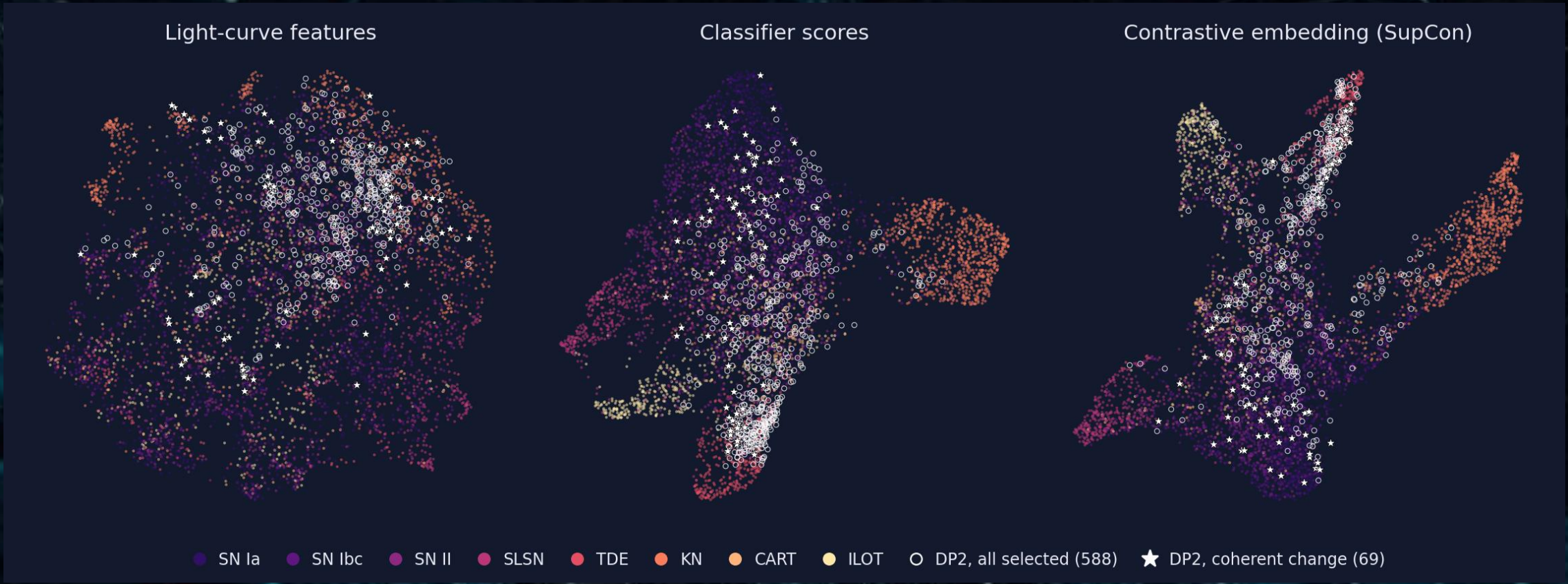
every participant who wants to sign; each team decides what it contributes

Check first

the Rubin terms for publishing DP2 results: aggregate numbers and figures, no catalogue rows

Next for this project: the wide survey area, labels from spectra, a model trained at the DP2 cadence.

Backup: the simulated map



A real transient has 4 real objects among its 10 nearest neighbours; chance gives 1.

Backup: learning the metric

Contrastive embedding (simulation)

0.48 to 0.60

nearest-neighbour balanced accuracy on simulated test objects, plain features to learned embedding (trees: 0.65)

21% to 55%

DP2 objects whose simulated neighbours agree with the classifier

4 groups

a Gaussian mixture on the embedding splits the 98 into groups that follow class and host position

Like / dislike head (69 likes)

AUC 0.89

on 407 DP2 objects the head never saw, all below the reliability cut (plain cosine similarity: 0.70)

24 of the top 50

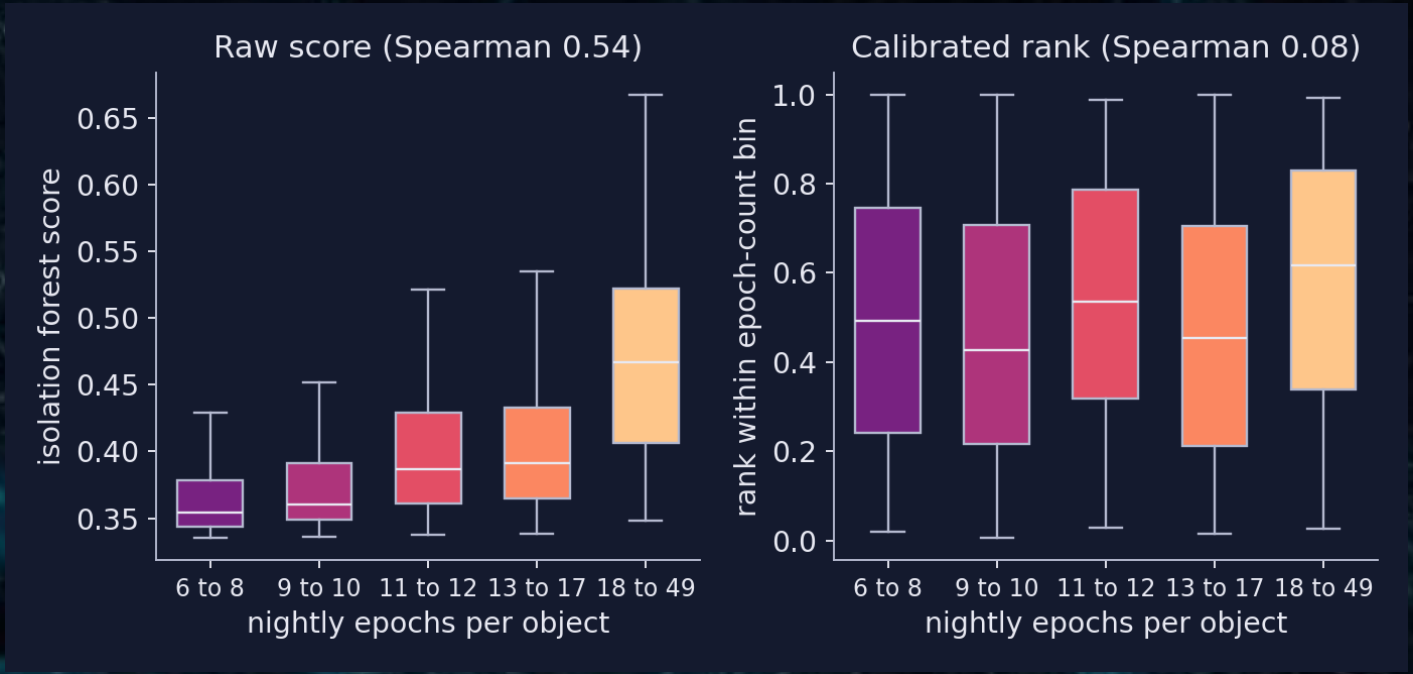
show a coherent change, against 7% in that sample

29 more transients

with a coherent change and no star at the position, recovered below the reliability cut: 69 becomes 98

A learned metric makes real neighbours meaningful; a few likes transfer to new objects.

Backup: outliers and their neighbours



89 outliers

top 15% of the calibrated rank: 75 in 4 families, 12 singletons

15 outliers

among the 98 coherent transients: 10 in 2 families, 5 singletons

more_like_this(id)

returns the nearest real transients with their class, set size and rank

The raw score follows the number of epochs; the rank within epoch-count bins does not. Most outliers have similar outliers next to them.

Most outliers have similar outliers next to them.

Backup: the code, step by step

1. train_sassafras6.py

simulation only: 329 six-band features, LightGBM, conformal sets at 90% coverage

2. loose_dp2.py, stageB_dp2.py

DP2 through TAP: catalogue cuts, forced photometry, quiet baseline, position in the Object table

3. apply_dp2.py

model on real transients: set sizes, domain classifier, anomaly rank, outlier families

4. supcon_dp2.py, userhead_dp2.py

contrastive embedding from the simulation; like/dislike head that recovers 29 transients

5. gmm_dp2.py, fast_dp2.py

Gaussian mixture on the embedding; search for events that fade by 1 mag within 6 days

6. reproduce.py

prints the numbers of this talk from the cached files, no TAP queries

Shared code: features6.py (same features for simulated and real light curves) and tap.py.